

# Connecting complex conductivity spectra to mercury porosimetry of sedimentary rocks

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## ABSTRACT

The measured spectral-induced polarization (SIP) response of highly porous granular materials can be modeled by upscaling the Nernst-Planck equation for a single grain and performing a convolution product with the particle size distribution. The distribution of relaxation times associated with the polarization of consolidated porous media seems, intuitively, to be controlled by the pore size distribution because of the radius of curvature of the interface between the solid and the pore water. We have explored a mechanistic model connecting the SIP response of consolidated media to mercury injection capillary porosimetry (MICP) data. First, we developed a mechanistic electric double layer (EDL) polarization model for porous media, which considered the polarization of the Stern layer, which was possibly enhanced by the polarization of the diffuse layer and the effect of the pore size distribution. The new model revealed that the

relaxation time of porous media such as sandstones could be overestimated if the enhancement of the overall polarization by the diffuse layer was not considered. Second, we have developed an empirical method to obtain the pore size distribution from MICP data by assuming a linear relation between the pore size and the pore throat size using a pore aspect ratio. Then, we compared the measured quadrature conductivities of nine samples with the theoretical quadrature conductivity obtained by convoluting the pore size distribution with the Warburg model. The Warburg model is considered to represent the quadrature conductivity response of a consolidated porous material characterized by an infinitively narrow unimodal pore size distribution coated by an EDL. The position of the peak of the phase was well-reproduced from the MICP data. In addition, the calculated broadness of the phase spectra was usually well-reproduced by convoluting the Warburg function with the pore size distribution.

## INTRODUCTION

The complex conductivity (or induced polarization) of water-saturated porous media, such as soils and rocks, describes their ability to store reversible electric charges at low frequencies (<10 kHz) like a capacitance (e.g., Vinegar and Waxman, 1984; Kemna et al., 2012). The measurement of complex conductivity can be performed either in the frequency domain (spectral-induced polarization, SIP) or in the time domain. In the past few decades, induced polarization has been shown to be a very promising geophysical method in hydrogeophysics for a broad number of applications including permeability estimation (Börner et al., 1996; Binley et al., 2005; Hördt et al., 2007; Revil and Florsch, 2010; Attwa and Günther, 2013; Weller et al., 2015), the characterization of near-surface lithology (Gazoty et al., 2012b), the detection of microbial growth in

porous media (Davis et al., 2006; Revil et al., 2012a), landfill characterization (Fianduca et al., 2012; Gazoty et al., 2012a), and the delineation of contaminant plumes (Vanhala, 1997; Chen et al., 2012; Deceuster and Kaufmann, 2012; Schwartz and Furman, 2012). Recent reviews regarding the use of this method can be found in Kemna et al. (2012) and Revil et al. (2012b).

Much research has been done to understand the fundamental mechanisms controlling induced polarization. Two mechanisms seem to contribute to the overall measured response below 10 kHz:

- 1) The first mechanism is related to the polarization of the Stern and diffuse layers (collectively known as the electric double layer, EDL) (see Schwarz, 1962; Schurr, 1964; Fixman, 1980; Lyklema et al., 1983; Leroy et al., 2008; Revil, 2012). Diffuse-layer polarization is usually considered to be smaller

Manuscript received by the Editor 31 January 2015; revised manuscript received 12 July 2015; published online 8 December 2015.

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than the contribution of the Stern layer (de Lima and Sharma, 1992; Lesmes and Morgan, 2001; Revil, 2012).

- 2) The second mechanism is called membrane polarization. Hittorf numbers represent the fraction of current carried by a single type of ionic charge carrier. Membrane polarization is caused by the change in the Hittorf numbers between the pores and the pore throats (Marshall and Madden, 1959; Vinegar and Waxman, 1984; Titov et al., 2002; Bückner and Hördt, 2013). We still do not know if this contribution is really important, but recent works seem to indicate that it may not be required to explain the measured SIP of porous rocks (Revil, 2012, 2013a, 2013b).

When the frequency increases to approximately 10 kHz, the Maxwell-Wagner polarization (or interfacial polarization) is significant and may control the observed polarization response (e.g., Leroy et al., 2008). Maxwell-Wagner polarization is caused by the formation of field-induced free charge distributions near the interface between the phases of the porous materials or colloidal suspensions (Maxwell, 1892; Wagner, 1914). Models based on the effective medium approximation (e.g., Hanai, 1960; Sen et al., 1981; Leroy et al., 2008) are efficient in modeling this contribution.

In a porous material, there can be several contributors to induced polarization, such as those associated with the presence of bacteria (Davis et al., 2006; Zhang et al., 2014) and dispersed semiconductors (Aal et al., 2014). In addition, heavy metals can strongly impact the observed polarization through their sorption in the Stern layer (Vaudelet et al., 2011a, 2011b). In the present work, we consider, therefore, a consolidated rock free of these spurious effects. Several studies have been already devoted to the modeling of the macroscopic induced polarization response of rocks (e.g., Ballarín et al., 1976; de Lima and Sharma, 1992). However, these works do not consider the distribution of the grain or pore sizes, and therefore they cannot model the observed broad distribution of relaxation times. Some improvements have been done more recently to solve this problem. For example, Lesmes and Morgan (2001) were the first to model the SIP responses of sedimentary rocks by combining Stern and diffuse-layer polarizations around solid particles (e.g., Schwarz, 1962; Fixman, 1980) with the Maxwell-Wagner polarizations discussed above. In their approach, the EDL polarization model for a single grain was convolved with the normalized probability density function associated with the particle size distribution. This approach was followed later by Leroy et al. (2008) and Leroy and Revil (2009) for glass beads and clays, respectively.

In the absence of dispersed semiconductors and bacteria in the porous material, and for supporting 1:1 pore water electrolytes (such as NaCl and KCl), the relaxation time distribution of consolidated porous materials has also been observed to be more directly related to the pore size distribution rather than the particle size distribution (Scott and Barker, 2003; Binley et al., 2005; Kruschwitz et al., 2010; Titov et al., 2010; Revil et al., 2012c). Therefore, it seems more meaningful to use the pore (throat) size rather than grain size data, while relating the microscale

EDL polarization to the measured SIP response of porous media. To our knowledge, a fundamental model for this behavior is still lacking.

The key in the upscaling process is the relationship between the pore size and the relaxation time. The EDL polarization theory for a single pore embedded in a rock matrix is lacking, and thus no theoretical equation to date is available to connect the pore size to the relaxation time of porous media. In practice, researchers use an empirical relation modified from the work of Schwarz (1962) by replacing the particle size with the pore size (Florsch et al., 2014; Revil et al., 2014), although no theoretical foundations have been provided in these works. To solve this problem, the first objective of this study is to connect the pore size to the relaxation time using a mechanistic polarization model of the EDL coating a water-filled pore embedded into a solid mineral phase.

Another difficulty in the upscaling process is how to appropriately obtain the pore size distribution information of the porous material. The current practice is to use the so-called pore throat size distribution interpreted from mercury injection capillary porosimetry (MICP) data. Nuclear magnetic resonance relaxation time distributions could offer an alternative approach. However, MICP data actually contain information about the pore size and pore throat size (Wardlaw, 1976; Tsakiroglou and Payatakes, 2000). Considering that the EDL polarization of porous media is mainly controlled by the pore size (e.g., Revil et al., 2012c, 2014), it seems necessary to separate the pore size information from the MICP data to model the SIP response of porous media.

## THEORY OF EDL POLARIZATION IN POROUS MEDIA

### Electric double layer

We first review the basic concepts pertaining to our understanding of the EDL in porous media. We consider a silicate rock satu-

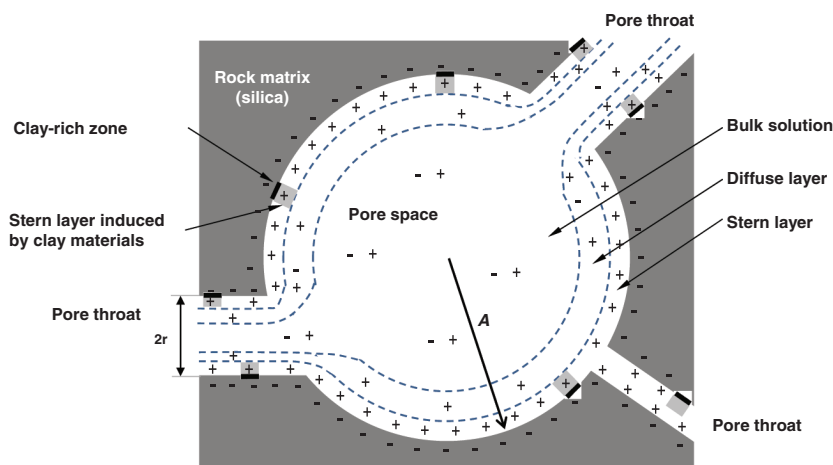


Figure 1. Sketch of the EDL in a spherical pore of radius  $A$  connected to other pores by three pore throats. The EDL consists of the Stern and diffuse layers. Some clay-rich zones (in light gray) are usually found on the surface of the rock matrix, and the resulted Stern layer may impede the movement of ions because of the low mobility of the counterions. Therefore, the Stern layer can be treated as discontinuous and will polarize when an alternating electric field is applied. Polarization of the Stern layer creates some polarization in the adjacent diffuse layer, which can enhance the overall polarization. The mean value of the ratio between the pore size and the pore throat size is called the aspect ratio of the pore space.

rated with water (Figure 1). The pore is connected to the remaining pore space by a set of pore throats. The radius of a pore  $A$  is assumed to be larger than the radii of the pore throats  $a$  that are connected to it. The surface of the solid (mineral) phase is considered to be negatively charged. This negative charge density is counterbalanced by a positive charge density in the Stern and diffuse layers (Figure 1).

### Stern layer polarization

We first assume that the EDL polarization is associated with the counterions bound to the surface of the silicate, and we neglect the exchange of ions between the Stern and diffuse layers and between the diffuse layer and the bulk pore water solution. This implies that there is no normal flux of ions in the Stern layer.

We consider a spherical pore, such as that depicted in Figure 1. An alternating (harmonic) electric field  $\mathbf{E} = \mathbf{E}_0 e^{i\omega t}$  (where  $i = \sqrt{-1}$  and  $\omega$  is the angular frequency) is applied to this pore. In the following, we use the spherical coordinates  $r$  and  $\theta$  with  $r = 0$  at the center of the pore and  $\theta = 0$  in the direction of the electric field. After applying the alternating electric field, the counterions in the Stern layer will deform tangentially, creating a charge density deviation within the Stern layer  $\Delta Q_\beta$ , which can be expressed as

$$Q_\beta = Q_\beta^0 + \Delta Q_\beta, \quad (1)$$

where  $Q_\beta^0$  is the original surface density of the charge carriers in the Stern layer (in  $\text{Mol m}^{-2}$ ), and  $Q_\beta$  is the charge density of the charge carrier in the Stern layer after applying the electric field. Note that  $\Delta Q_\beta$  is a function of  $\theta$  and time  $t$ . Because the counterions can only move tangentially, the conservation of ions in the Stern layer can be locally expressed as

$$\frac{\partial Q_\beta}{\partial t} = \nabla \cdot \mathbf{I}_b, \quad (2)$$

where  $\mathbf{I}_b$  denotes the ionic flux density in the Stern layer ( $\text{Mol m}^{-2} \text{s}^{-1}$ ). The flux density  $\mathbf{I}_b$  actually has two components, one induced by the applied electric field and the second arising from diffusion within the Stern layer:

$$\mathbf{I}_b = \frac{D_{(+)}^s}{A} \left( \frac{eN Q_\beta}{RT} \frac{\partial \varphi_b}{\partial \theta} + \frac{\partial \Delta Q_\beta}{\partial \theta} \right), \quad (3)$$

where  $D_{(+)}^s$  is the diffusion coefficient for ions in the Stern layer ( $\text{m}^2 \text{s}^{-1}$ ),  $\varphi_b$  is the electric potential (in volts) in the Stern layer after applying the electric field,  $A$  is the pore radius (in meters),  $e$  is the elementary charge ( $1.6 \times 10^{-19} \text{ C}$ ), and  $T$  is the temperature (in Kelvin). The other two parameters appearing in equation 3,  $N$  and  $R$ , denote the Avogadro number ( $6.02 \times 10^{23} \text{ Mol}^{-1}$ ) and the universal gas constant ( $8.3145 \text{ J K}^{-1} \text{ Mol}^{-1}$ ), respectively.

We assume that the applied field is sufficiently weak, so the problem can be linearized. Therefore, equation 2 can be simplified as

$$\frac{\partial Q_\beta}{\partial t} = i\omega \Delta Q_\beta = \nabla \cdot \mathbf{I}_b. \quad (4)$$

Inserting equation 3 into equation 4 results in the following partial differential equation:

$$i\omega \Delta Q_\beta = \frac{D_{(+)}^s}{A^2} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left[ \sin \theta \left( \frac{eN Q_\beta}{RT} \frac{\partial \varphi_b}{\partial \theta} + \frac{\partial \Delta Q_\beta}{\partial \theta} \right) \right]. \quad (5)$$

Equation 5 is similar to equation 15 in Schwarz (1962), who considers a solid spherical particle immersed in an electrolyte. Here, however, the particle is replaced by a pore, and therefore the grain size is replaced by the pore size. Equation 5 is solved by Schwarz (1962). His solution for  $\Delta Q_\beta$  can be expressed as

$$\Delta Q_\beta = -\frac{eN Q_\beta / RT}{1 + i\omega \tau_s} \Delta \varphi_b, \quad (6)$$

where the time constant  $\tau_s$  is given by

$$\tau_s = \frac{A^2}{2D_{(+)}^s}. \quad (7)$$

We consider a simple supporting 1:1 electrolyte, such as KCl or NaCl. The total current density  $\mathbf{j}_s$  ( $\text{A m}^{-2}$ ) associated with the ionic flux density  $\mathbf{I}_b$  of the counterions ( $\text{K}^+$  or  $\text{Na}^+$ ) is

$$\mathbf{j}_s = e \mathbf{I}_b. \quad (8)$$

Considering equations 3, 6, and 8 and assuming  $\varphi_b^0 = \varphi_b$  and  $\partial \varphi_b / \partial \theta \cong \partial \Delta \varphi_b / \partial \theta$  lead to

$$j_s = \frac{i\omega \tau_{bs}}{1 + i\omega \tau_{bs}} e^2 Q_\beta \beta_{(+)}^s \frac{\partial \varphi_b}{\partial \theta}, \quad (9)$$

where  $\beta_{(+)}^s$  is the mobility of the counterions in the Stern layer. The mobility of the counterions in Stern layer  $\beta_{(+)}^s$  can be related to the diffusion coefficient  $D_{(+)}^s$  through the Nernst-Einstein relationship  $D_{(+)}^s = \beta_{(+)}^s T k_b$  for a symmetric monovalent (1:1) electrolyte (such as KCl or NaCl;  $k_b$  denotes the Boltzmann constant). Equation 9 is in fact a form of Ohm's law for electric conduction along the surface of the pores. It follows that the surface conductivity can be expressed as the following complex-valued function:

$$\sigma_{\text{EDL}}^* = \frac{i\omega \tau_s}{1 + i\omega \tau_s} e^2 Q_\beta \beta_{(+)}^s. \quad (10)$$

This contribution can be added to the contribution of the bulk pore water (which is frequency independent and represented by  $\sigma_w$ ) and upscaled to a representative elementary volume (see, for instance, Revil [2013a], who used a volume averaging technique). We will not need, however, such a model in the present paper because we are only interested in the distribution of the relaxation time.

### Role of the diffuse layer

In our previous papers, the polarization of the diffuse layer was neglected during the upscaling of the local equations to the scale of a granular material (Leroy et al., 2008; Revil, 2013a, 2013b; Revil et al., 2014). It is, however, possible that the exchange of ions between the diffuse layer and the Stern layer occurs during Stern layer polarization, and therefore, a more complete treatment is needed. The equivalent surface charge density for the whole diffuse layer

is written as  $Q_s^0$  at equilibrium (Figure 1). After applying the alternating electric field  $Q_s^0$  changes to  $Q_s$ , and these two parameters are related by

$$Q_s = Q_s^0 + \Delta Q_s, \quad (11)$$

where  $\Delta Q_s$  is the change in the surface charge density in the diffuse layer. To keep electroneutrality of the EDL,  $\Delta Q_s$  and  $\Delta Q_\beta$  have the following relationship:

$$\Delta Q_s = -\Delta Q_\beta. \quad (12)$$

We define the potential at the boundary between the Stern layer and the diffuse layer as  $\varphi_d^0$  before applying the alternating field. This potential becomes  $\varphi_d$  after the application of the electric field. The difference between  $\varphi_d^0$  and  $\varphi_d$  is written as  $\Delta\varphi_d$ . This potential change  $\Delta\varphi_d$  is also a function of  $\theta$  and  $t$ , and it can be related to the density change in the diffuse layer  $\Delta Q_s$  by introducing the differential capacitance (Lyklema et al., 1983):

$$\Delta\varphi_d = -\Delta Q_s / C_d, \quad (13)$$

$$C_d = \varepsilon_a \varepsilon_0 \kappa \cosh\left(\frac{eN\varphi_d^0}{2RT}\right), \quad (14)$$

for a 1:1 electrolyte. The parameter  $\varepsilon_a$  denotes the relative dielectric constant of the bulk solution,  $\varepsilon_0$  denotes the permittivity of vacuum, and  $\kappa^{-1}$  denotes the Debye length:

$$\kappa^{-1} = \sqrt{\frac{\varepsilon_a \varepsilon_0 RT}{2e^2 N^2 C_f}}, \quad (15)$$

where  $C_f$  is the electrolyte concentration (salinity,  $\text{mol L}^{-1}$ ).

We define the electric potential at the outer boundary of the EDL as  $\varphi_a$ , and we have

$$\varphi_b = \varphi_a + \varphi_d. \quad (16)$$

Differentiating equation 16 with respect to  $\theta$ , and using equations 12 and 13 lead to the following partial differential equation:

$$\begin{aligned} \frac{\partial\varphi_b}{\partial\theta} &= \frac{\partial\varphi_a}{\partial\theta} + \frac{\partial\varphi_d}{\partial\theta} = \frac{\partial\varphi_a}{\partial\theta} + \frac{\partial\Delta\varphi_d}{\partial\theta} = \frac{\partial\varphi_a}{\partial\theta} - \frac{1}{C_d} \frac{\partial\Delta Q_s}{\partial\theta} \\ &= \frac{\partial\varphi_a}{\partial\theta} + \frac{1}{C_d} \frac{\partial\Delta Q_\beta}{\partial\theta}. \end{aligned} \quad (17)$$

Equation 17 illustrates that the polarization of bound ions,  $\partial\varphi_b/\partial\theta$ , is influenced by the tangential oscillations of the potential of the diffuse layer,  $\partial\Delta\varphi_d/\partial\theta$ .

Substituting equation 17 into equation 5 gives

$$i\omega\Delta Q_\beta = \frac{D_{(+)}^s}{A^2} \frac{eNM}{\sin\theta} \frac{\partial}{\partial\theta} \left[ \sin\theta \left( \frac{eNQ_\beta^0}{RT} \frac{1}{M} \frac{\partial\varphi_a}{\partial\theta} + \frac{\partial\Delta Q_\beta}{\partial\theta} \right) \right], \quad (18)$$

where

$$\begin{aligned} M &= 1 + \frac{eNQ_\beta}{RTC_d} = \frac{\kappa Q_\beta}{2eNC_f \sinh\left(\frac{eNQ_\beta^0}{2RT}\right)} \\ &= 1 + \frac{2}{\cosh\left(\frac{eNQ_\beta^0}{2RT}\right)} \left[ \frac{Q_0\kappa}{4eNC_f} - \sinh\left(\frac{eNQ_\beta^0}{2RT}\right) \right], \end{aligned} \quad (19)$$

and where  $Q_0$  denotes the surface charge density of the mineral surface. This surface charge density is connected to  $Q_\beta$  and  $Q_s$  through a global electroneutrality relationship (e.g., Leroy et al., 2008):

$$Q_0 + Q_\beta + Q_s = 0. \quad (20)$$

Compared with equation 5, equation 18 involves the additional contribution from the diffuse layer by considering the exchange of ions between the Stern and diffuse layers. Following the same approach used in Schwarz (1962), we can derive the following solution based on equation 18:

$$\Delta Q_\beta = -\frac{eNQ_\beta^0/RT}{1 + i\omega\tau_{sd}} \frac{\varphi_a}{M}, \quad (21)$$

and

$$\tau_{sd} = \frac{A^2}{2MD_{(+)}^s}. \quad (22)$$

We can rewrite equation 10 as a frequency-dependent surface conductivity:

$$\sigma_{\text{EDL}}^* = \frac{i\omega\tau_{sd}}{1 + i\omega\tau_{sd}} e^2 Q_\beta \beta_{(+)}^s. \quad (23)$$

Equations 22 and 23 describe the electric conductivity induced by the polarization of the Stern and diffuse layers. The parameter  $M$  can be treated as a coefficient that corrects the relaxation time of the Stern polarization from the polarization of the diffuse layer. The upper and lower bounds for this coefficient will be discussed in the next section. We can, here again, upscale this surface conductivity using a bulk conductivity  $\sigma_w$  inside the pores following the approach obtained by Revil (2013a).

### The correction coefficient $M$

The direction of the polarization field induced by the EDL  $\mathbf{E}^*$  is opposite to the (external) applied electric field  $\mathbf{E}$ , and therefore  $\mathbf{E}^*$  will reduce the amplitude of the shifts of the bound counterions. Consequently, the polarization of the diffuse layer actually reduces the relaxation time of the bound ions because of the exchange of ions between diffuse and Stern layers. This implies that the value of  $M$  should be equal to or larger than one. In addition, the value of  $M$  is also affected by the surface charge density  $Q_0$ , the charge density for the Stern layer  $Q_\beta$ , and the salinity of the electrolyte  $C_f$ . To analyze the range of values taken by the correction coefficient  $M$ , we use equation 19 to test its dependence on the charge density  $Q_\beta$  and the salinity  $C_f$ . We also need to connect the potential at the

boundary between the Stern layer and diffuse layer  $\varphi_d^0$  with the charge density  $Q_S$ . This can be achieved by the Gouy-Chapman relationship to describe the equivalent surface charge density of the diffuse layer (Gouy, 1910; Chapman, 1913):

$$Q_S = -\sqrt{8\epsilon_d\epsilon_0k_bTC_f} \sinh\left(\frac{e\varphi_d^0}{2k_bT}\right). \quad (24)$$

The numerical results are shown in Figure 2. The parameter  $M$  ranges between 1 and 3. According to equation 22,  $M > 1$  indicates that the polarization of the diffuse layer reduces the (observed) relaxation time. Here,  $M$  equals one when the contribution of the diffuse-layer polarization is negligible. This corresponds to the case characterized by an extremely low surface charge density (on the order of  $10^{-4} \text{ C m}^{-2}$ ; see Figure 2), which is, for instance, the case of carbonate rocks as discussed further below. As the charge density of the diffuse layer  $Q_S$  increases, the value of  $M$  increases as well up to an upper limit of three. This upper bound is reached when the value of  $Q_S$  exceeds  $0.01 \text{ C m}^{-2}$  for a salinity of  $C_f = 0.0001 \text{ mol L}^{-1}$ . Figure 2 shows that the value of  $M$  is significantly affected by the salinity of the electrolyte (at low salinities,  $M$  equals one, whereas  $M$  equals three at high salinity). When the salinity of the electrolyte increases, the calculated curve of  $M$  is shifted toward the direction of a high surface charge density, implying a smaller contribution from the polarization of the electric diffuse layer.

For clays, it has been shown that the average surface charge density is approximately two elementary charges per  $\text{nm}^2$ , i.e.,  $0.32 \text{ C m}^{-2}$  (see the discussions in Revil, 2012), and that approximately 45%–99% of the counterions are within the Stern layer (Leroy et al., 2008; Revil, 2012). This means that the surface charge density of the diffuse layer is typically in the range

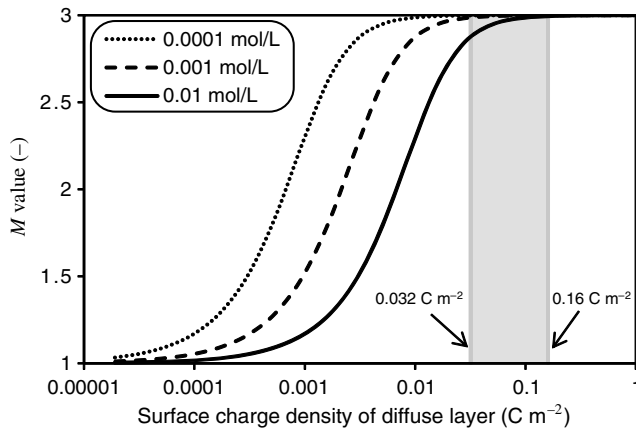


Figure 2. Correction coefficient  $M$  as a function of the surface charge density of the diffuse layer. This coefficient is associated with the polarization of the diffuse layer with  $M = 1$  implying no polarization of the diffuse layer, whereas  $M = 3$  is the maximum value reached by  $M$  at high surface charge densities. Three electrolyte salinity values were used in the calculation: 0.0001, 0.001, and  $0.01 \text{ mol L}^{-1}$ . The surface charge density of silica was assumed as two elementary charges per  $\text{nm}^2$  (i.e.,  $0.32 \text{ C m}^{-2}$ ) (e.g., see discussions in Revil, 2012). For quartz and clay minerals, approximately 45%–99% of the total counterions are localized within the Stern layer (see Leroy et al., 2008 for silica and Leroy and Revil, 2009 and Revil, 2012 for clays). These values correspond to the shaded area in the figure.

$0.16\text{--}0.032 \text{ C m}^{-2}$ , which corresponds to the shaded area in Figure 2. The associated  $M$  value in the shaded area is approximately three, indicating that for clays, the contribution of the diffuse-layer polarization would significantly enhance the polarization of the Stern layer. In this case, the relaxation time is expected to be three times smaller than the case for which only Stern layer polarization is considered. For silica, the presence of the diffuse layer would have a similar effect regarding the shift of the time constant due to the existence of the diffuse layer.

As discussed briefly above, the value of  $M$  for carbonates is expected to be close to one because their surface charge density is significantly lower than for silicates and aluminosilicates (e.g., Pokrovsky et al., 1999). Indeed, carbonate minerals impose to the pore water a pH and a pCa that are close to their isoelectric point. As a consequence, the charge density at the surface of the grains is very small with respect to the charge density of silicates and aluminosilicates.

In summary, our model is consistent with our previous Stern polarization models (Revil, 2013a, 2013b; Revil et al., 2014). The presence of a diffuse layer adjacent to the Stern layer creates an amplification of the overall polarization. This amplification can be accounted for by using the correction factor  $M$ , which is equal to one in the absence of a diffuse layer, and it can reach a maximum value of three for a well-developed diffuse layer.

## PORE SIZE AND COLE-COLE RELAXATION TIME

Several studies suggest that the permeability of porous media can be estimated through induced polarization measurements (e.g., Binley et al., 2005; Revil and Florsch, 2010). To improve the permeability prediction from SIP measurements, Revil et al. (2012c) propose the following relation between the conductivity Cole-Cole relaxation time  $\tau_{cc}$  and the characteristic pore size  $\Lambda$ :

$$\tau_{cc} = \frac{\Lambda^2}{2D_{(+)}^s}, \quad (25)$$

where the parameter  $\Lambda$  is a measure of the radius of the pore throats (Johnson et al., 1987; Avellaneda and Torquato, 1991; Revil and Cathles, 1999). Equation 25 is an empirical equation that was derived by Revil et al. (2012c) using dimensional analysis. It should be noted that this relationship is mainly based on the Stern layer polarization theory (i.e., equation 7), and the influence of the diffuse layer is not considered.

To account for the contribution of the diffuse layer, we now modify equation 25 by adding the effect of the correction coefficient  $M$ . A new equation is obtained connecting the Cole-Cole relaxation time  $\tau_{cc}$  to the characteristic pore size  $\Lambda$ :

$$\tau_{cc} = \frac{\Lambda^2}{2MD_{(+)}^s}. \quad (26)$$

Published data are used to check the proposed relationship (Figure 3). The parameter  $\Lambda$  in Figure 3 is determined from the permeability  $K$  (in  $\text{m}^2$ ) and the intrinsic formation factor  $F$  using  $\Lambda^2 = 8FK$  (Avellaneda and Torquato, 1991). Theoretical curves calculated from equation 26 with  $M = 1$  and 3 are also plotted in Figure 3. Compared with the theoretical curve with  $M = 1$

(i.e., Stern layer polarization only), the curve with  $M = 3$  (i.e., the Stern layer and diffuse-layer polarizations) gives a better fit to the experimental data. The root-mean standard deviations (RMSDs) between the measured and predicated  $\log_{10} \tau_{cc}$  are 0.56 and 0.32 for the theoretical curves with  $M = 1$  and  $M = 3$ , respectively. It is clear that neglecting the influence of the diffuse layer could cause errors if we try to determine the length scale  $\Lambda$  from SIP measurements. Note that the Cole-Cole relaxation time can also be determined from time-domain induced polarization measurements in field conditions (e.g., [Fiandaca et al., 2012](#)) and laboratory conditions (e.g., [Tarasov and Titov, 2007](#)).

## COMPLEX CONDUCTIVITY OF A POROUS ROCK

### Cole-Cole model

The [Cole and Cole \(1941\)](#) model is often used to describe the complex electric conductivity of porous media. It can be expressed as (e.g., [Tarasov and Titov, 2013](#))

$$\sigma^*(\omega) = \sigma_{\infty} - \frac{M_n}{1 + (i\omega\tau_{cc})^c}, \quad (27)$$

where the normalized chargeability  $M_n$  ( $\text{S m}^{-1}$ ) is defined as  $M_n = \sigma_{\infty} - \sigma_0$ ,  $\tau_{cc}$  is the Cole-Cole relaxation time (in seconds),  $c$  (dimensionless) is the Cole-Cole exponent ( $0 < c \leq 1$ ), and  $\sigma_{\infty}$  and  $\sigma_0$  denote the low- and high-frequency asymptotic limits of the electric conductivity, respectively. We do not need specific relationships for these asymptotic limits, but for the interested reader,

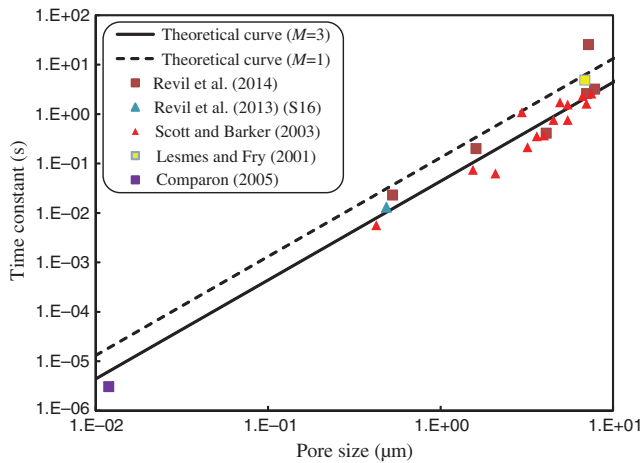


Figure 3. Relationship between the characteristic pore throat size and the Cole-Cole time constant for different porous media. The dashed and solid lines in the figure correspond to theoretical curves (equation 26) with  $M = 1$  (i.e., Stern layer polarization) and  $M = 3$  (i.e., the Stern layer and diffuse-layer polarizations), respectively. The experimental data are collected from [Revil et al. \(2014, 2013\)](#), [Scott and Barker \(2003\)](#), [Lesmes and Morgan \(2001\)](#), [Comparon \(2005\)](#). The RMSDs between the measured and predicated  $\log_{10} \tau_{cc}$  are 0.56 and 0.32 for the curves with  $M = 1$  and  $M = 3$ , respectively. Note that the time constant corresponding to the data from [Scott and Barker \(2003\)](#) was obtained from the peak frequency of the phase. For porous media characterized by small chargeabilities, the two time constants are expected to be very close to each other ([Tarasov and Titov, 2013](#)).

we have derived rigorous expressions in some of our previous papers (see [Revil, 2013a, 2013b](#); [Revil et al., 2014](#)). In the following, we will consider that the characteristic time constant  $\tau_{cc}$  is not dependent on the salinity. This assumption is generally supported by experimental data (see [Joseph et al., 2015](#)). When  $c = 0.5$ , equation 27 is called a Warburg transfer function, whereas when  $c = 1$ , equation 27 is called a Debye transfer function. In this paper, we will use a Warburg model rather than a Debye model while conducting the calculation. The reasons for this choice are extensively discussed by [Revil et al. \(2014\)](#). It is imposed by the underlying physics of the problem (leaking versus nonleaking capacitance) and by the observations. Indeed, the spectrum for a porous material characterized by a unimodal and narrow pore size distribution can be fitted by a Warburg model, not by a Debye model.

### Relation between relaxation time distribution and pore size distribution

If the porous medium can be characterized by a pore size distribution, the effective electric conductivity induced by the EDL can be obtained by the convolution of the pore size distribution with the conductivity response obtained for a delta distribution of the pore size (Green's function) according to the superposition principle ([Lesmes and Morgan, 2001](#); [Leroy et al., 2008](#)):

$$\sigma_{EDL}^*(\omega) = f(A) \otimes \sigma_{EDL}^*(A, \omega), \quad (28)$$

where  $\otimes$  is the convolution product and  $A$  is the pore radius. The pore size probability density function  $f(A)$  is normalized:

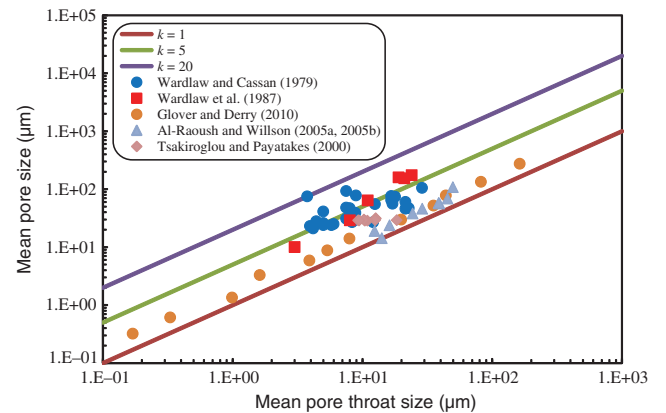


Figure 4. Relationship between the mean pore throat size and the mean pore size for different porous media. The three solid lines denote a linear relationship (see equation 34), with  $k = 1, 5$ , and  $20$ , respectively. The pore size data in [Wardlaw and Cassan \(1979\)](#) and [Wardlaw et al. \(1987\)](#) are made by direct observation of thin sections and resin casts of the rock samples. The pore size data in [Glover and Dery \(2010\)](#) are calculated using the technique in [Glover and Walker \(2009\)](#). The pore size and pore throat size data in [Al-Raoush and Willson \(2005a, 2005b\)](#) are extracted from X-ray tomography. The pore size and pore throat size in [Tsakiroglou and Payatakes \(2000\)](#) are extracted from mercury injection porosimetry data.

$$\int_0^{\infty} f(A) dA = 1. \quad (29)$$

Using the convolution theorem based on the superposition principle, the complex electric conductivity of a porous medium with pore size distribution  $f(A)$  can be expressed as (Revil et al., 2014)

$$\sigma_{\text{EDL}}^*(\omega) = \sigma_{\infty} - M_n \int_0^{\infty} \frac{h(\tau)}{1 + (i\omega\tau_{\text{cc}})^c} d\tau, \quad (30)$$

where  $c = 0.5$  for a Warburg transfer function taken as the Green's function of the problem. The distribution  $h(\tau)$  is also a normalized probability density function obeying

$$\int_0^{\infty} h(\tau) d\tau = 1. \quad (31)$$

Pore size  $A$  is related to relaxation time  $\tau$  through equation 22. The pore size distribution  $f(A)$  is associated with the relaxation time distribution  $h(\tau)$ . Their relationship can be obtained following the same type of analysis made by (Revil and Florsch, 2010)

$$h(\tau) = \frac{MD_{(+)}^s}{F_h} \frac{f(A)}{A^2}, \quad (32)$$

where

$$F_h = \int_0^{+\infty} f(A) d \ln A. \quad (33)$$

Equation 32 can be used to calculate the relaxation time distribution based on the pore size distribution, and by considering equations 22 and 30, the effective conductivity induced by EDL polarization can

then be calculated. Note that the pore size distribution can also be constrained from the distribution of relaxation times that are inverted from SIP data. This alternative approach has been explored

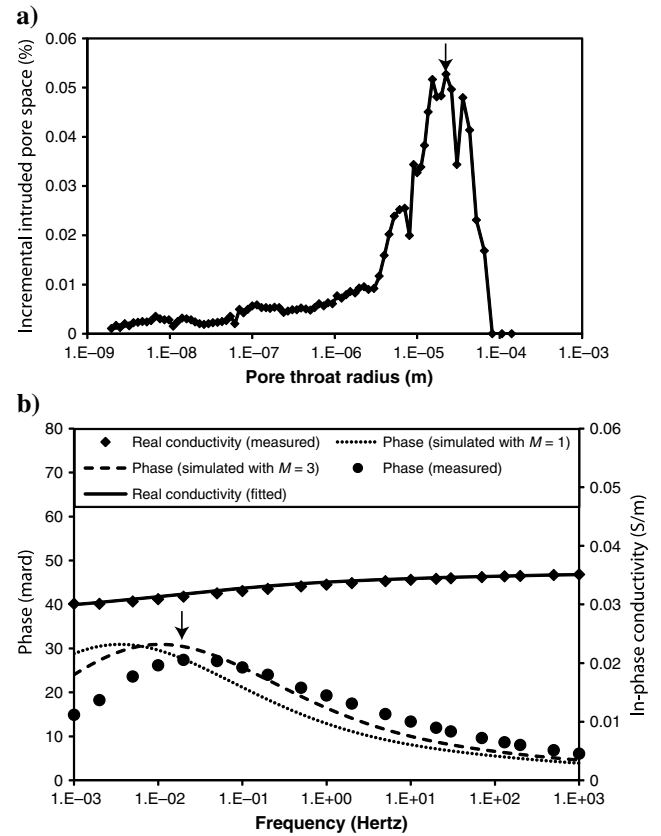


Figure 5. Properties of sample S490 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of the dominant peak frequency.

**Table 1. Properties of the samples used in this study: porosity  $\phi$ , permeability  $K$ , formation factor  $F$ , pore size  $\Lambda = (8KF)^{1/2}$ , Cole-Cole relaxation time  $\tau_0$ , and pore throat radius at peak of the distribution  $r_c$  (peak).**

| Sample                     | $\phi(-)$ | $K$ (mD) | $F$ (-) | $\Lambda$ (m) | $\tau_0$ (s) | $r_c$ (peak) ( $\mu\text{m}$ ) |
|----------------------------|-----------|----------|---------|---------------|--------------|--------------------------------|
| S499                       | 0.265     | 1103     | 5.6     | 7.0           | 2.6          | 22                             |
| S498                       | 0.206     | 35.9     | 9.0     | 1.6           | 0.20         | 4.5                            |
| S490                       | 0.233     | 635      | 12.1    | 7.8           | 3.2          | 25                             |
| S493                       | 0.232     | 115      | 18.3    | 4.1           | 0.41         | 6                              |
| S439                       | 0.208     | 2.62     | 13.3    | 0.53          | 0.023        | 0.9                            |
| S436                       | 0.306     | 1623     | 4.0     | 7.2           | 25.5         | 55                             |
| Rothbach sandstone         | 0.166     | —        | —       | —             | 2.22         | 9.8                            |
| Schleerith sandstone       | 0.163     | —        | —       | —             | 0.80         | 1.2                            |
| Thuringian Basin sandstone | 0.145     | —        | —       | —             | 1.86         | 3.7                            |
| Wetterstein limestone      | 0.0033    | —        | —       | —             | 10.87        | 0.5                            |

in several papers (e.g., Lesmes and Morgan, 2001; Florsch et al., 2014; Revil et al., 2014).

### Relation between pore size and pore throat size

The MICP measurement is a standard technique that can provide information about the pore system of porous media. Although MICP data are usually interpreted in the terms of pore throat size distribution, they actually contain volume information about the pores and pore throats (Wardlaw, 1976). To relate pore size  $A$  to pore throat size  $r$ , a “pore-to-throat radius ratio”  $k$  ( $\geq 1$ ) (also called the aspect ratio or constriction factor of the pores; see Erlich et al. [1991]) is introduced

$$k = A / r. \quad (34)$$

This concept is introduced by Owen (1952) to model the electric resistivity of rocks (see Jorden and Campbell, 1986). This parameter also plays a critical role in capillary trapping in carbonate and sandstones during two-phase flow of immiscible fluids and to explain the occurrence of Haines jumps (e.g., Oh and Slattery, 1976). The value of  $k$  is usually in the range of 5–10 for quartz-cemented sandstones, bioturbated sandstones, and sucrosic dolomites (see

Hartmann and Beaumont, 1999). That said, for a given sandstone, the value of  $k$  can take a broad range of values (e.g., Lindquist et al., 2000). Experimental data about the mean value of the pore-to-throat-radius ratio of different rocks are collected from the literature and displayed in Figure 4. The value of  $k$  ranges from one to five with a maximum value of  $k = 20$ . Tanino and Blunt (2012) show that for sandstones, the value of  $k$  is generally between one and four.

For porous media with  $k = 1$ , i.e., the pore throat size is equal to the pore size, the pore system can be represented by a bundle of capillary tubes with different sizes. We also assume that when the associated pore/pore throat size distribution from MICP testing is unimodal, the pore and pore throat have the same size; i.e.,  $k = 1$ . Values of  $k$  larger than one imply that the pore system needs to be characterized by two different pore systems, i.e., pore and pore throat. We further assume the size distribution of the pore and pore throat follows the same distribution, although the respective volume fractions are different. Based on these assumptions, the pore and pore throat can be separated and we can obtain the pore size distribution from MICP data. It should be noted that the pore/pore throat size distribution is usually bimodal for porous media with  $k \geq 1$ .

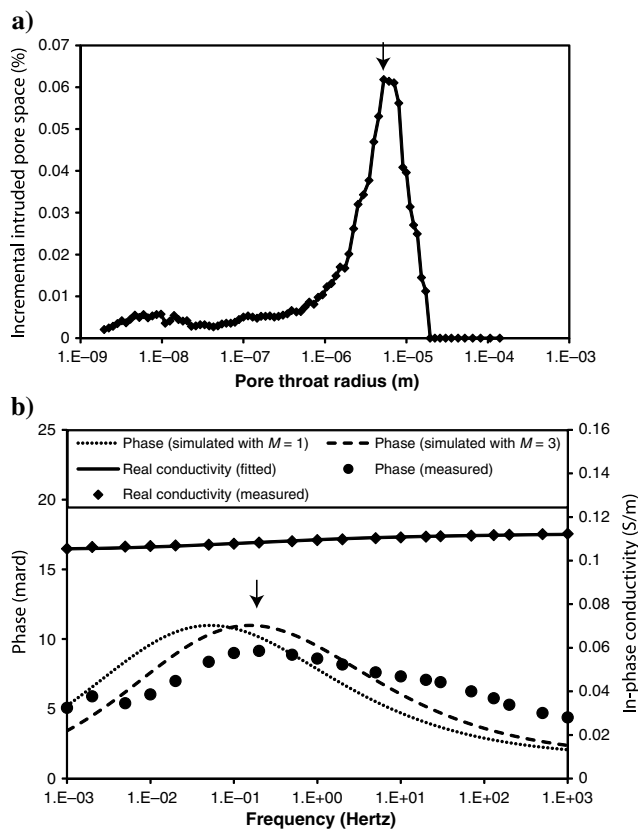


Figure 6. Properties of sample S493 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

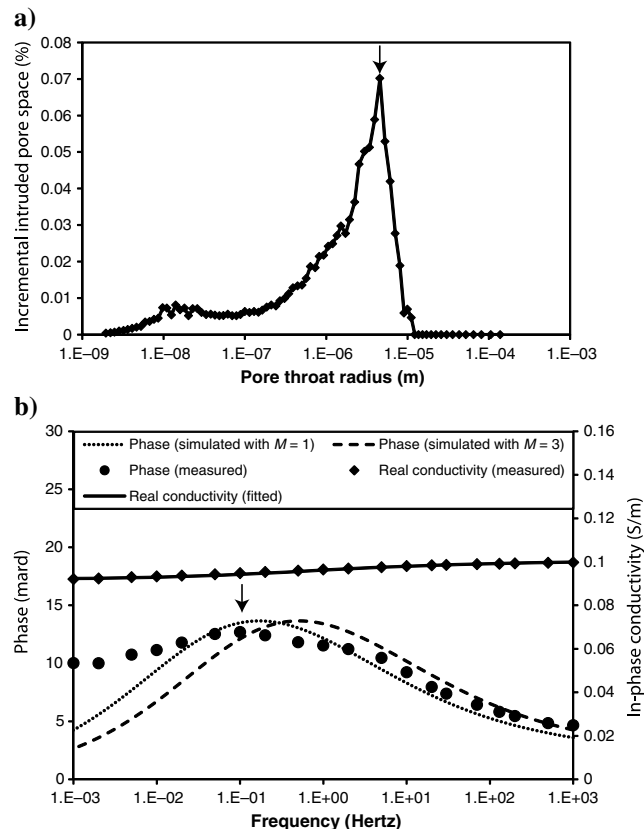


Figure 7. Properties of sample S498 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

## RESULTS

In this section, we calculate the SIP responses from the MICP data for 10 samples, and the calculations are compared with the measurements. The petrophysical properties of the samples are provided in Table 1. Most of the samples are sandstones, and their porosity ranges from 0.145 to 0.265. There is also a limestone (the Wetterstein limestone), which is characterized by low porosity (approximately 0.0033). Note that samples S499, S498, S490, S493, S439, and S436 are from Revil et al. (2014), and the permeability and intrinsic formation factor are available (see Table 1). The pore size  $\Lambda$  is also available as determined as  $(8KF)^{1/2}$ . The other four samples are newly measured (A. Kemna, personal communication, 2015), and their permeability, formation factor, and pore size  $\Lambda$  are not available. The complex conductivities of all samples were measured using the four-terminal method with a high-precision impedance analyzer (Zimmermann et al., 2008) and nonpolarized silver-silver chloride (Ag-AgCl) electrodes. The measured resistance was converted into resistivity/conductivity using a geometric factor that was numerically calculated based on the geometry of the sample and the position of the electrodes.

The MICP data of all the samples are shown in Figures 5a, 6a, 7a, 8a, 9a, 10a, 11a, 12a, 13a, and 15a, respectively. These data do not

exhibit a “perfect” unimodal pore throat distribution. It seems that there is always (at least) a second peak in addition to the main peak. For example, the incremental intruded pore space of sample S493 has two peaks. Although the position for the main peak is at  $r = 6 \mu\text{m}$ , the second peak is at  $r = 0.008 \mu\text{m}$ . In our calculation, as long as the incremental intruded pore space for the second peak is less than the half of the main peak of a sample, we classify it as the unimodal type. Otherwise, we classify the sample as the bimodal type. For the bimodal samples, the position of the second peak is picked as the pore size with the second highest incremental intruded pore space. According to the above-mentioned criteria, seven of our samples are classified as unimodal pore/pore throat size distribution. Three of them, including one limestone, are classified as bimodal pore/pore size distribution.

## Sandstones with unimodal pore/pore throat size distribution

Once the relationship between pore size and pore throat size is known, the MICP data can be interpreted in terms of pore (not pore throat) size distribution. Using equations 30 and 32, the effective conductivity induced by the EDL can then be calculated. For unimodal pore/pore throat size distribution,  $k$  can be treated as one. In

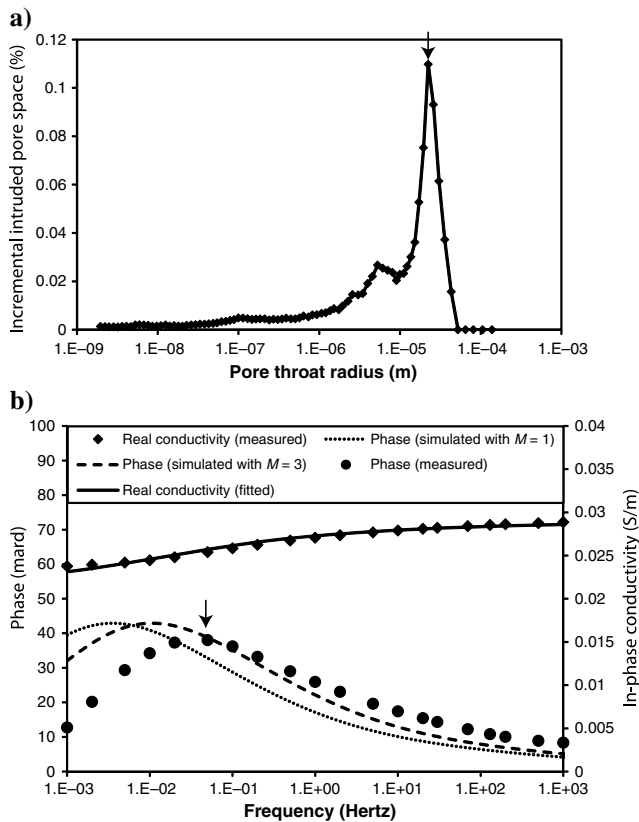


Figure 8. Properties of sample S499 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of the dominant peak frequency.

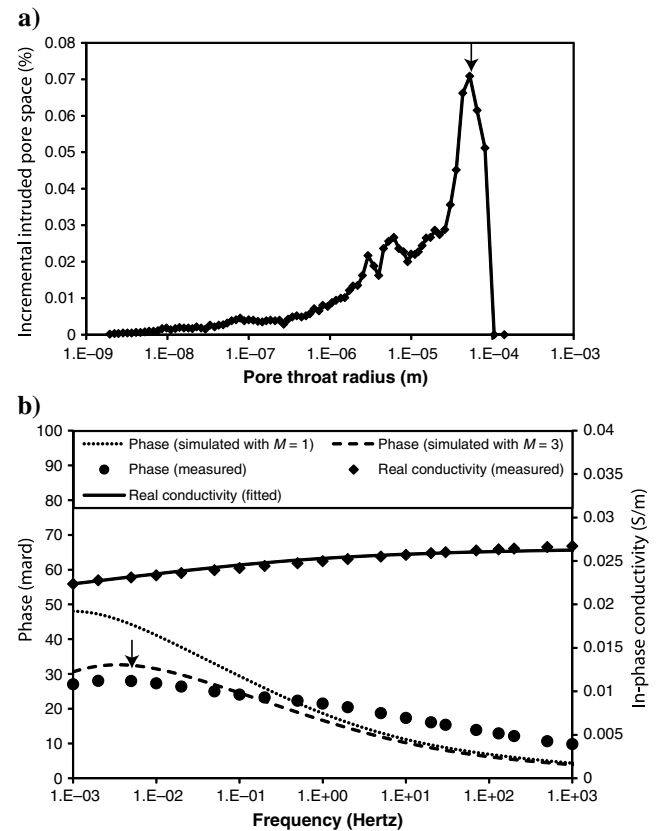


Figure 9. Properties of sample S436 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of the dominant peak frequency.

this subsection, eight sandstone samples with the MICP and SIP data are used to verify the previous model. The first six samples are from the study of Revil et al. (2014), and the other data are new.

The calculated complex conductivity is shown in Figures 5–11 together with the experimental data. Concerning our experimental data, the error from the phase was determined using three cycles per frequency. At low frequencies, most of the uncertainty comes from the lack of stability of the electrodes, whereas at high frequencies, the errors are coming from capacitive and inductive couplings. As explained in several of our previous papers, great care was taken to minimize these errors (see, for instance, Revil and Skold, 2011). We use a threshold of 10% to determine the quality of the phase measurements. If the standard deviation is higher than 10%, we further stacked the data until the standard deviation was smaller than 10%. The data with a standard deviation higher than 10% were disregarded.

In the calculation, the parameters  $\sigma_\infty$  and  $M_n$  in equation 30 are determined by fitting the in-phase conductivity to the experimental data. Although these parameters can be determined using interfacial electrochemical properties of porous media, such as the cation exchange capacity or the surface charge density (e.g., Lesmes and Morgan, 2001; Leroy and Revil, 2009), we did not follow the same

approach in this study mainly because there is not enough information about the surface chemical properties of the samples, such as salinity and pH values.

The calculated results in Figures 5–11 consider two polarization mechanisms. Although  $M = 1$  corresponds to the Stern layer polarization only,  $M = 3$  represents the enhanced polarization associated with the diffuse layer. The measured SIP data are generally in better agreement with the calculated SIP responses with  $M = 3$  than with  $M = 1$ . For instance, in Figure 11, the phase response of a Rothbach sandstone, of the peak and the width, is well-reproduced based on Hg porosimetry data by considering  $M = 3$  (see the dashed line). In contrast, the calculated phase curve with  $M = 1$  (dotted line in Figure 11) deviates significantly from the measurement: Although the width of the calculated phase seems to agree fairly well with the measurement, the peak position is underestimated. In conclusion, the results in Figures 5–11 clearly show that, for sandstones with a unimodal pore/pore throat size distribution, the SIP responses, especially the relaxation time, can be reasonably well-reproduced from the MICP data using our proposed methodology, although the shape of the phase is not necessarily well-modeled. The shape of the phase is mainly controlled by the parameter  $c$  in the Cole-Cole model (fixed and considered to be equal to one-half in this test).

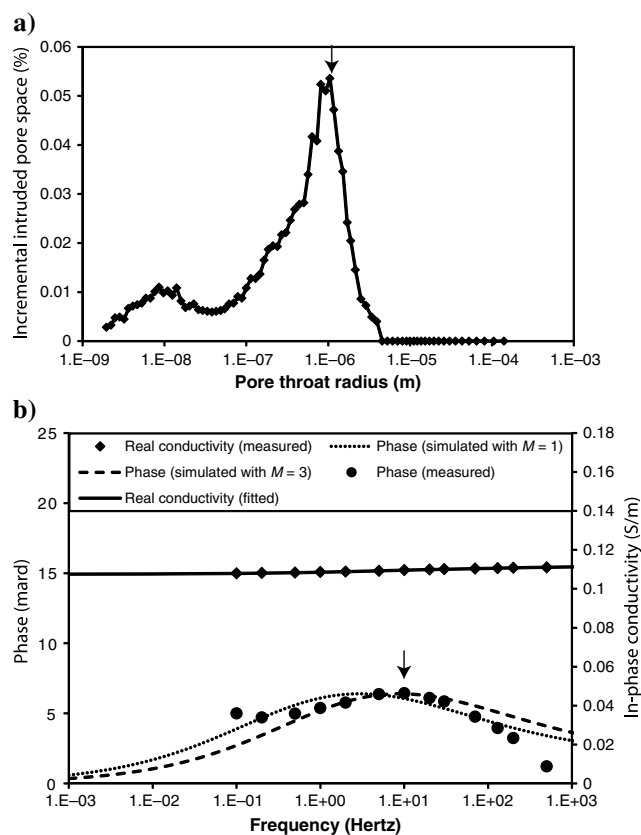


Figure 10. Properties of sample S439 (from Revil et al., 2014). (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

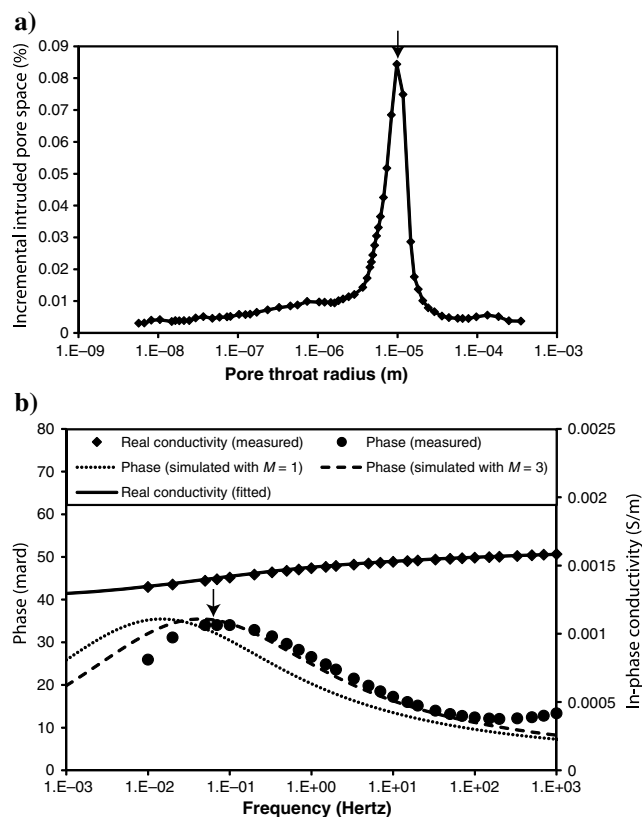


Figure 11. Properties of a Rothbach sandstone. (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. Dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

### Sandstones with bimodal pore/pore throat size distribution

For sandstones with bimodal pore/pore throat size distributions, we believe the pore system is composed of several pores and pore throats. It is assumed that the linear relationship between the pore size and pore throat size (equation 34 with  $k > 1$ ) holds for the samples here. As stated, once the value of  $k$  is determined, the SIP responses can be calculated using equations 30 and 32 based on the MICP data. It is difficult to theoretically determine the  $k$  value based on the measured bimodal pore/pore throat size distribution because we have no information about the volume fractions of either the pore or the pore throat.

To tackle this problem, we propose an empirical approach to determine the value of  $k$  from the measured bimodal pore/pore throat size distribution. We first identify the two peak positions of the pore/pore throat from the pore/pore throat size distribution. For example, the two radii corresponding to the two peaks in Figure 12a are 1.23 and 0.048  $\mu\text{m}$ , respectively. Then, the  $k$  value can be calculated as the square root of the ratio between these two radii, for example, as 5.05 for the Schleerieth sandstone data shown in Figure 12. This choice is purely arbitrary and was obtained by trial and error. It is clear that it will need to be (1) confirmed with additional data sets and (2) justified with more direct observations.

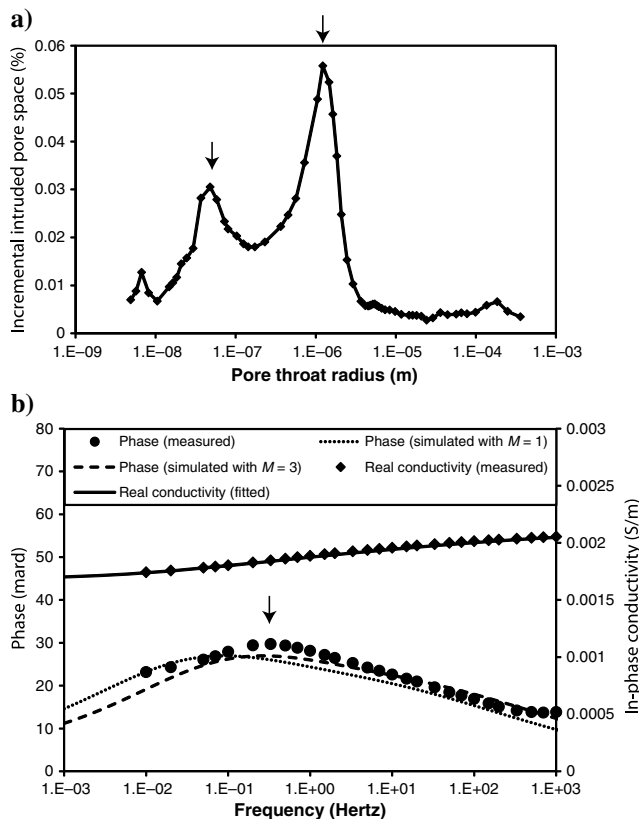


Figure 12. Properties of the Schleerieth sandstone. (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

The values of  $k$  of two sandstones (Schleerieth and Thuringian Basin sandstones) with bimodal pore size distribution (Figures 12 and 13) are calculated based on the proposed empirical method. The SIP responses are then calculated according to the method mentioned previously, and the results are presented in Figures 12 and 13. Similarly,  $M = 1$  and  $M = 3$  were used in the calculation to analyze the influence of diffuse-layer polarization on the simulated SIP responses. Figures 12 and 13 show that the shapes of the induced polarization spectra of the samples can be fairly reproduced based on the pore/pore throat size distribution data using either  $M = 1$  or  $M = 3$ . This indicates that the shape/broadness of the phase is mainly controlled by the pore/pore throat size distribution. For the position of peak frequency, however, calculations with  $M = 1$  and 3 give different values. Although the simulated peak frequencies (or relaxation times) of the two samples with  $M = 3$  are in good agreement with the measurements, results with  $M = 1$  significantly underestimate the peak frequencies. The results in Figures 12 and 13 once again imply that the Stern and diffuse-layer polarization should be considered in modeling the SIP responses of sandstones.

The results in Figures 5–13 show a fair agreement between the measured and calculated SIP spectra of porous media. It should be noted that all the samples in this study have relatively large pores ( $>1 \mu\text{m}$ ). For samples with smaller pores and lower permeability, further studies are still needed to test the validation of the method.

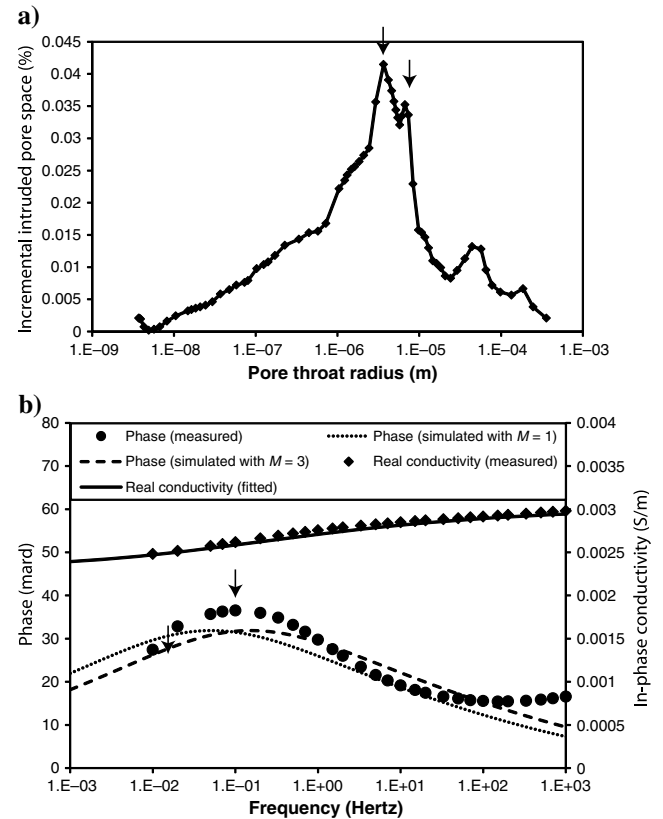


Figure 13. Properties of the Thuringian Basin sandstone. (a) Pore/pore throat size distribution data from mercury injection porosimetry. The arrow denotes the position of dominant pore throat size. (b) SIP responses. The dashed and dotted lines in the figure correspond to phase angles calculated with  $M = 3$  and  $M = 1$ , respectively. The arrow denotes the position of dominant peak frequency.

This method might fail because the diffusion length scale is not controlled by pore sizes in samples with a pore size smaller than  $1\ \mu\text{m}$ .

To better illustrate the influence of diffuse-layer polarization in modeling the SIP responses, in Figure 14 we plot the measured relaxation times against the calculated relaxation times for all the samples shown in Figures 5–13. Although only the Stern layer polarization is considered in Figure 14b, the calculation shown in Figure 14a considers the combined Stern and diffuse-layer polarization processes. As shown in Figure 14, it is obvious that considering diffuse-layer polarization (i.e., using  $M = 3$ ) significantly increases the accuracy of the calculated relaxation time when compared with calculations with  $M = 1$ . The associated RMSDs between measured and calculated  $\log_{10} \tau$  are 0.34 and 0.56 for Figure 14a and 14b, respectively.

### Limestone with bimodal pore size distribution

Limestone is composed largely of calcite, and the surface charge density of calcite in an electrolyte is substantially lower at thermodynamic equilibrium than for siliciclastic materials (e.g., Pokrovsky et al., 1999). As shown in Figure 2, when the surface charge density of the diffuse layer is lower than  $0.0001\ \text{C m}^{-2}$ , the value of  $M$  is expected to be close to one. Therefore, the influence of diffuse-layer polarization can be ignored. To check this point, we calculated the

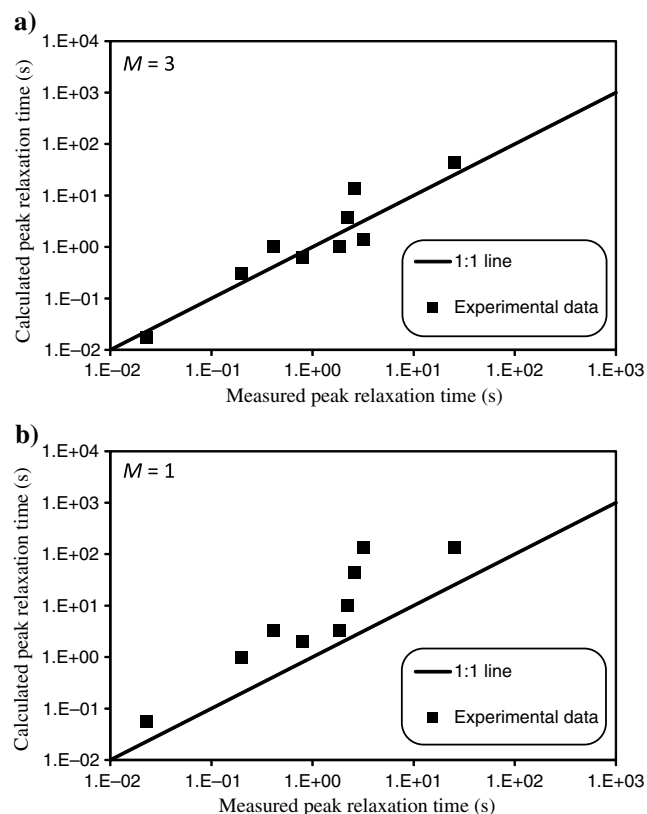


Figure 14. Relaxation times of the nine siliciclastic core samples. (a) Calculated relaxation time based on the Stern layer polarization amplified by the diffuse-layer polarization (i.e.,  $M = 3$ ). (b) Calculated relaxation time is based on only Stern layer polarization (i.e.,  $M = 1$ ). Note that the errors bars determined from the uncertainties in picking the peak of the pore size and the peak of the phase are smaller than the size of the symbols.

SIP responses of the Wetterstein limestone based on the measured bimodal pore/pore throat size distribution shown in Figure 15a. Similarly, the calculated real conductivity curve was fitted to the measurement to obtain the parameters  $\sigma_\infty$  and  $M_n$ , and the  $k$  value was determined using the empirical method from MICP data. In the calculation, the value of  $M$  was assumed as one to account for the low surface charge density of limestone. The results are shown in Figure 15b, and it is found that the calculated peak frequency of the Wetterstein limestone is in good agreement with the measurement, indicating the validation of the above analysis. Although the shapes of the calculated and measured phase deviate from each other in Figure 15b, in the following subsection, we will show that the deviation in shapes can be accommodated by changing the Cole-Cole exponent.

### Cole-Cole exponent $c$

As shown above, the SIP response of sedimentary rocks can be modeled well from MICP data by considering the combined Stern- and diffuse-layer polarization mechanisms. However, the shapes of the phase sometimes cannot be reproduced using a Warburg model, for example, for the data shown in Figures 6, 9, 12, 13, and 15. To explore the influence of the Cole-Cole exponent  $c$  in the calculation of the SIP responses, we now fit the value of  $c$  by fitting the calculated phase curve to the measurement. The samples in Figures 6, 9, 12, 13, and 15 were recalculated. The results are shown in Figure 16.

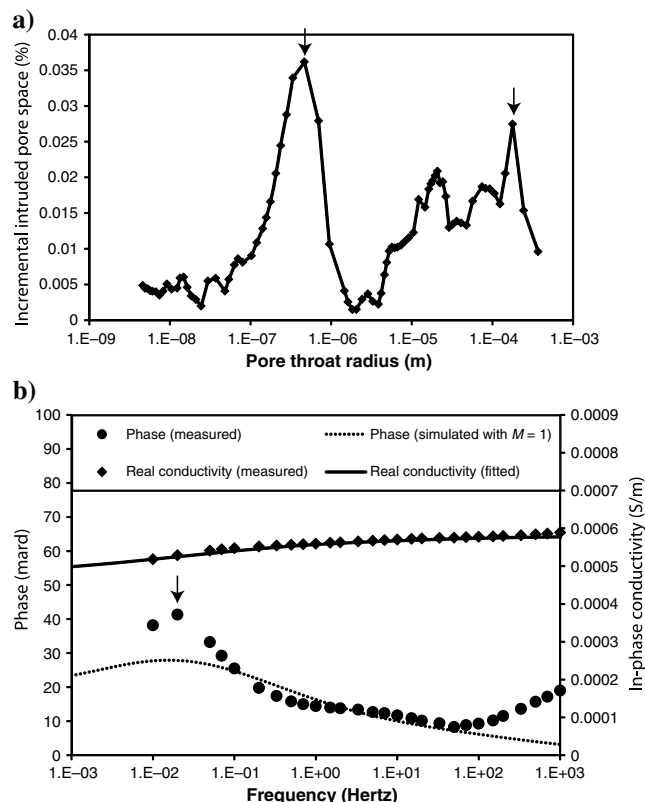


Figure 15. The properties of the Wetterstein limestone. (a) Pore/pore throat size distribution data from mercury injection porosimetry. (b) SIP responses. The dotted line in the figure corresponds to phase angles calculated with  $M = 1$ . The arrows denote the position of dominant pore throat size and peak frequency.

We found that the calculated SIP responses are in better agreement with the measurement compared with the cases with  $c = 0.5$ . In these examples, the relaxation time is still determined by the convolution of the pore size distribution. This implies that the Cole-Cole exponent is actually not a constant and it differs among rocks. Only in the case of well-sorted, sandstones with a narrow unimodal pore throat size distribution, should the exponent approach the theoretical value ( $c = 0.5$  for the Warburg-model; see the discussion in [Revil et al., 2014](#)). A broader pore size distribution would cause a flattening of the phase spectrum with the Cole-Cole exponent  $c$  get-

ting smaller (Figure 16). The Cole-Cole exponent  $c$  is therefore strongly influenced by the broadness of the pore size distribution. A broad pore size distribution can be caused by the fractal behavior of the pore space as discussed recently by [Zhang and Weller \(2014\)](#) (see also [Revil, 2012](#)). A very broad pore size distribution should finally result in a flat phase spectrum. [Weller et al. \(2015\)](#) show in their study that only 21 of 56 samples exhibit a clear relaxation peak in their phase. Most of the other samples show a flat (constant phase angle) behavior or a steadily increasing phase shift (see also [Vinegar and Waxman, 1984](#)). It should also be noted that the polarization

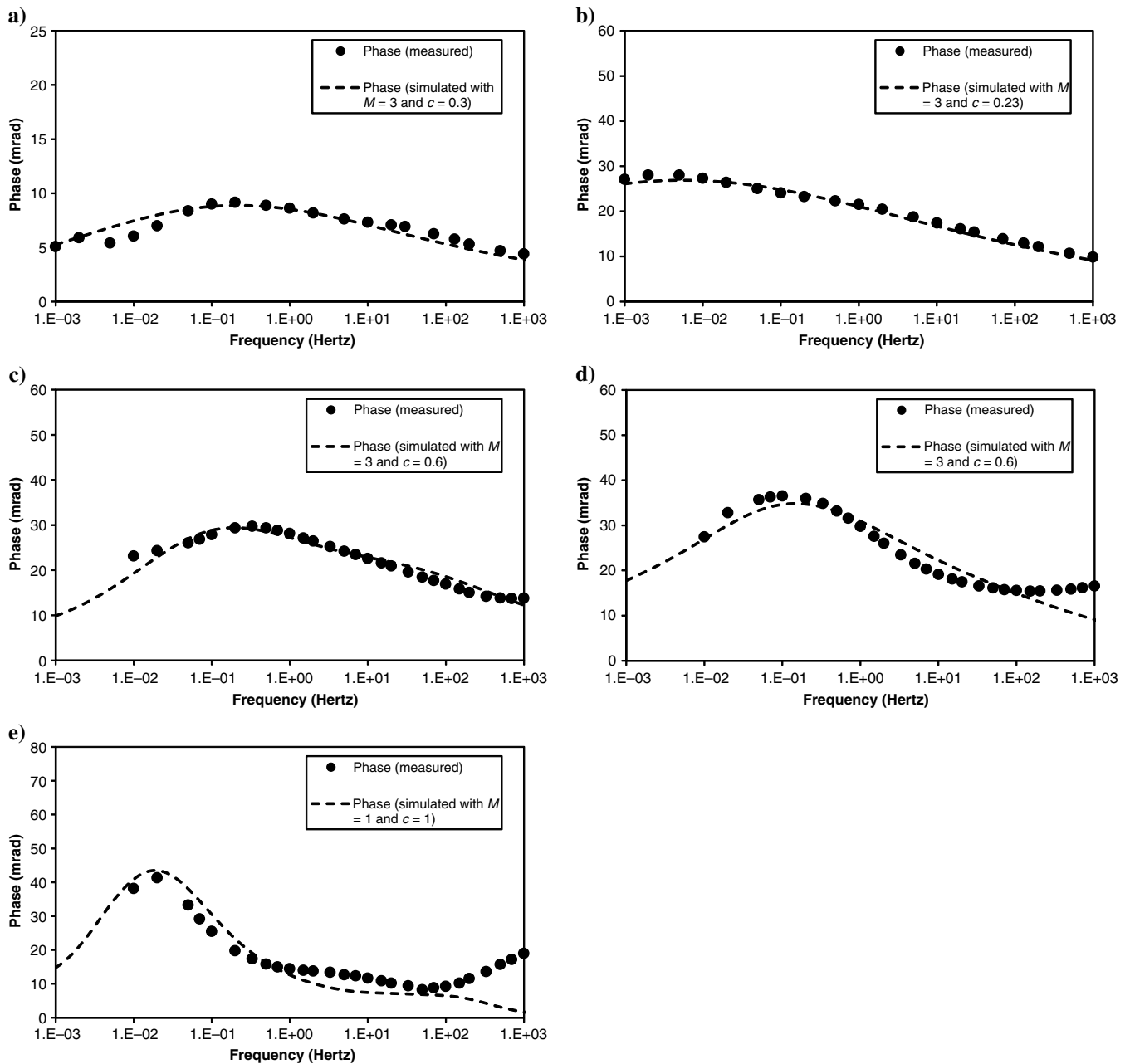


Figure 16. The calculated and measured phase shift with the Cole-Cole exponent as a free parameter. (a) Sample S493 (see Figure 6). (b) Sample S436 (see Figure 9). (c) Schleerieth sandstone (see Figure 12). (d) Thuringian Basin sandstone (see Figure 13). (e) Wetterstein limestone (see Figure 15). Note that all figures are shown with the same range of frequencies covering six decades.

peak may be hidden by the high-frequency slope caused by the Maxwell-Wagner polarization or is located outside of the investigated frequency interval. In future research works, it will therefore be necessary to analyze the variation of  $c$  value in different sedimentary rocks, such that the  $c$  value can be appropriately estimated, e.g., from the width of the pore size distribution.

### Limitation of the present approach

Our study shows that a fair agreement between the measured and modeled phase spectra is possible for samples with relatively large pores ( $>1\ \mu\text{m}$ ), where a clear relation between the pore diameter and the relaxation time  $\tau$  has been shown to exist. The present method may fail for porous media characterized by small pore radii ( $<1\ \mu\text{m}$ ) and low permeability ( $<0.1\ \text{mD}$ ). Indeed, in this case, the diffusion length scale may not be controlled by the pore diameter. Kruschwitz et al. (2010) document that a reliable relation between the time constant and modal pore throat radius ( $D_0$ ) exists only for samples with  $D_0 > 5\ \mu\text{m}$ . Therefore, further tests will be required to find the exact validity of the present method for small pore sizes.

### CONCLUSION

We have first developed an EDL polarization model for a water-filled pore embedded in the rock matrix. We have demonstrated that the presence of a diffuse layer can increase the polarization of the Stern layer, and a correction coefficient (comprised between one and three) is introduced in the Stern layer polarization theory. The polarization of the diffuse layer is related to the exchange of ions between the Stern layer and diffuse layers. It does occur during Stern layer polarization when an alternating electric field is applied. The new model reveals that the relaxation time might be three times smaller than anticipated with Stern layer polarization alone in the case of sandstones characterized by high surface charge densities (approximately two elementary charges per  $\text{nm}^2$ ,  $0.32\ \text{C m}^{-2}$ ). For porous media characterized by low surface charge densities, such as limestones (approximately  $0.0001\ \text{C m}^{-2}$ ), the influence of the diffuse layer seems to be negligible. This result will need to be confirmed, however, by additional data sets.

Following the same idea, we modified the relation between the Cole-Cole relaxation time and the characteristic pore size, which was proposed to predict the permeability of porous media from SIP measurements. It was shown that the modified relation could better fit the measured data, which were collected from the literature. Published experimental data indicate that the radius of a pore is roughly proportional to the pore throat radius. This linear relationship can be used to obtain the pore size distribution from the MICP data. We have calculated SIP responses of several samples, including sandstones, one mudstone, and one limestone, by performing a convolution between the pore size distribution with the conductivity response obtained when all the pores have the same size. A comparison between the calculation and SIP data shows that the SIP response of sedimentary rocks, especially the position of the peak frequency, can be reproduced based on the MICP data. Although the calculated shape/broadness of the phase spectra sometimes may deviate from the measurements, the deviation can be accommodated by fitting the Cole-Cole exponent  $c$  in place of taking  $c = 0.5$  (the Warburg formula).

### ACKNOWLEDGMENTS

We thank A. Kemna for some of the data provided in this paper. We thank the Office of Science (BER), U.S. Department of Energy (award no. DE-FG02-08ER646559). We thank the associate editors, K. Keating, A. Binley, G. Fiandaca, and A. Weller for their very constructive reviews.

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